

Course Title	Applied Thermodynamics	Course Code	ME2002
Department	Department of Electrical Engineering (DEE)	Campus	Lahore
Knowledge Profile	Engineering in Society (WK7)	Credit Hrs.	3
Knowledge Area	Interdisciplinary Engineering (KA09)	Grading Scheme	Relative
HEC Knowledge Area	Inter Disciplinary Engg. Breadth (Elective)	Applicable From	Spring 2023
SDG	4 Quality Education		
Pre-requisite(s)			

Course Objective	This course gives an introduction of basic concepts of thermodynamics properties, first law and second law of thermodynamics, control mass and control volume approaches and their application to power and refrigeration cycles.
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No.	Assigned Program Learning Outcome (PLO)
02	An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
06	An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solution to complex engineering problems.
07	An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development

I = Introduction, R = Reinforcement, E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Analyze a thermodynamics process	A1, M1	C4	2
2	Analyze energy of a system using energy equation	Q1, M2	C4	7
3	Analyze modes of heat transfer	Q2, M2	C4	7
4	Apply Second Law of Thermodynamics to quantify the performance of a control mass and control volume thermodynamic system.	Q3, F	C3	7
5	Analyze the behavior of power plants based on the Rankine cycle, including the effect of enhancements such as superheat, reheat and regeneration.	A2, F	C4	6

Text Book(s)	Title	Fundamentals of Thermodynamics
	Author	Richard E. Sonntag, Claus Borgnakke and Gordon J. Van Wylen
	Publisher	John Wiley & Sons Inc., 8th Edition, 2012
Ref. Book(s)	Title	Fundamentals of Engineering Thermodynamics
	Author	Michael J. Moran and Howard N. Shapiro
	Publisher	John Wiley & Sons Inc., 6th Edition, 2000
	Title	Applied Thermodynamics for Engineering Technologists
	Author	T.D Eastop and A. McConkey
	Publisher	Pearson Education Ltd. 5th Edition, 1993

Week	Course Contents/Topics	Chapter*	CLO*
1	Introduction to thermodynamics. Energy transfer by heat work and mass.	1, 2	1
2	Introduction to thermodynamics, Properties of pure substances and Property Tables	2	1
3	T-v Diagram and p-v Diagram	2	1
4	The Ideal gas equation of state, compressibility factor.	2	1
5	Modes of Heat Transfer	3	3
6	First law of thermodynamics for control mass system.	3	2
7	First law of thermodynamics for control volume system.	3	2
8	Internal energy, enthalpy and specific heats of ideal gases.	3,4	2
9	The Ideal gas equation of state, its applications for first law of thermodynamics for close and open system.	4	2
10	Problem Solving and Revision	3,4	2
11	The second law of thermodynamics, its statement and application to heat engine and refrigeration cycles.	5	4
12	Entropy, calculation of entropy and its applications.	5,6	4
13	T-s relations, entropy change of liquid and solids.	6	4
14	Second law of thermodynamics for control volume	6	4
15	Introduction to Carnot and Rankine cycle	7	5
16	Problem Solving for Steam cycle.	5,6,7	5

***Reference book chapters are given in brackets**

Assessment Tools	Weightage
Quizzes(3), Assignments(3)	20.0%
Midterm (1+2)	30.0%
Final Exam	50.0%